

Linear Regression

$$y(x, w) = w_0 + w_1 x_1 + \dots + w_D x_D$$

Basis Functions

Keep the model linear in the weights, but allow non-linear transformations of the inputs.

$$y(x, w) = w_0 + \sum_{j=1}^{M-1} w_j \phi_j(x)$$

- Model is non-linear in x .
but, linear in weights.

$$y(x) = w^T \phi(x) \quad \text{where}$$

$$\phi(x) = (\phi_0(x), \phi_1(x), \dots, \phi_{M-1}(x))^T$$

Case 1: If x is a single scalar (1-D input) $M=2$

$$\text{Then } \phi(x) = [1 \ x \ x^2]^T \text{ and}$$

$$y = w_0 + w_1 x + w_2 x^2$$

Case 2: If x is a vector (D -dimensional input) $M=2$

$$\text{Let } x = (x_1, x_2, x_3, \dots, x_D)$$

$$(A) \ \phi_0(x) = 1$$

$$M=1] \ \phi_1(x) = x_1, \ \phi_2(x) = x_2, \dots, \phi_D(x) = x_D$$

$$M=2] \ \phi_{D+1}(x) = x_1^2, \ \phi_{D+2}(x) = x_2^2, \dots, \phi_{2D}(x) = x_D^2$$

$$y(x) = w_0 + \sum_{i=1}^D w_i x_i + \sum_{i=1}^D w_{D+i} x_i^2$$

